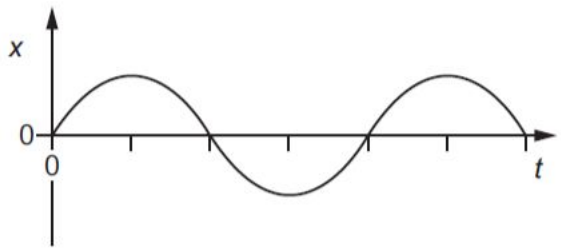
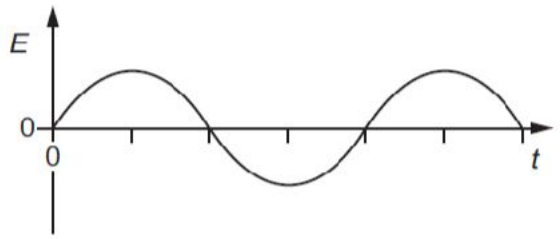
13 A body is undergoing undamped oscillations that are simple harmonic.

A graph of the variation of displacement $x$ of the body with time $t$ is shown.

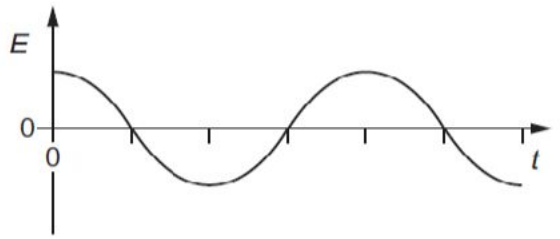



Which graph shows the variation of the kinetic energy $E$ of the body with time $t$?

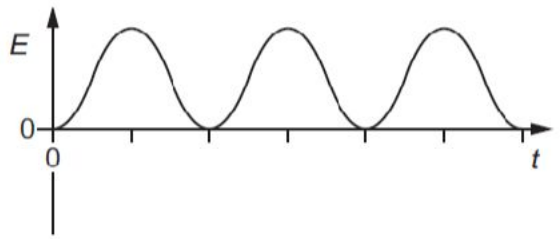
**A**



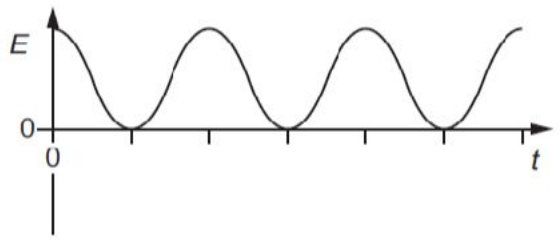

**B**




**C**




**D**




Ans: **D**

At $t = 0$, displacement of body is $x = 0$ at equilibrium position which means its velocity and hence KE is maximum. There are 2 cycles in a period. Hence option **D**.